

Project Title: Predicting Movie Success Based on Early Production Data

Abstract:

This project aims to predict whether a movie will be a box office success using data available before its release, such as budget, genre, director, cast, and production company. By building a predictive model, the project will help stakeholders make informed decisions early in the filmmaking process.

Introduction:

Movies are expensive to produce, and many factors influence their success. Being able to predict performance early could save millions. This project explores how data mining can be used to estimate a film's success at the box office using early-stage production features.

Literature Review:

Past studies have looked at various movie attributes and their correlation with revenue. IMDb, Box Office Mojo, and Kaggle datasets have been used in predictive modeling with mixed success. Some models use regression, while others use classification to determine whether a movie is a hit or a flop.

Methodology:

- Dataset: Publicly available datasets like IMDb, TMDb, or Kaggle's "The Movies Dataset."
- Tools: Python, pandas, scikit-learn, possibly XGBoost for classification.
- Techniques: Data cleaning, feature engineering, exploratory data analysis, classification modeling (logistic regression, decision tree, etc.).

Evaluation Plan:

- Metrics: Accuracy, F1 score, ROC-AUC.
- A movie will be labeled a "success" based on profitability or box office rank.

- Evaluate model performance using train/test split and cross-validation.

Timeline:

- Week 1-2: Data collection and cleaning
- Week 3: EDA and feature engineering
- Week 4-5: Modeling and evaluation
- Week 6: Final report and presentation slides

Sources:

- Kaggle: The Movies Dataset
- IMDb datasets
- Relevant academic papers on predictive modeling in the film industry